# Statistics: Questions and Full Solutions

## Question 01

**English**

The fifteen observations $1^2, 2^2, 3^2, \ldots, 15^2$ are grouped into the ten classes $[0, 25), [25, 50), [50, 75), \ldots, [225, 250)$, each class containing its left end point but not its right end point. Let $\bar{x}$ be the arithmetic mean of the fifteen observations, and let $\bar{x}_g$ be the arithmetic mean of the resulting continuous frequency distribution, computed with the class marks. Then $\bar{x}_g - \bar{x}$ is equal to

(A) $-\dfrac{7}{2}$

(B) $-\dfrac{1}{6}$

(C) $\dfrac{3}{2}$

(D) $\dfrac{5}{2}$

Only one option is correct.

**हिन्दी**

पंद्रह प्रेक्षणों $1^2, 2^2, 3^2, \ldots, 15^2$ को दस वर्गों $[0, 25), [25, 50), [50, 75), \ldots, [225, 250)$ में समूहित किया जाता है, जहाँ प्रत्येक वर्ग में उसका बायाँ अन्त्य बिन्दु सम्मिलित है परन्तु दायाँ अन्त्य बिन्दु नहीं। माना $\bar{x}$ इन पंद्रह प्रेक्षणों का समान्तर माध्य है तथा $\bar{x}_g$ प्राप्त सतत बारंबारता बंटन का वर्ग चिह्नों द्वारा परिकलित समान्तर माध्य है। तब $\bar{x}_g - \bar{x}$ बराबर है

(A) $-\dfrac{7}{2}$

(B) $-\dfrac{1}{6}$

(C) $\dfrac{3}{2}$

(D) $\dfrac{5}{2}$

केवल एक विकल्प सही है।

**Solution**

Actual mean: $\sum_{k=1}^{15} k^2 = \dfrac{15 \cdot 16 \cdot 31}{6} = 1240$, so $\bar{x} = \dfrac{1240}{15} = \dfrac{248}{3}$.

Grouping: $[0, 25)$ contains $1, 4, 9, 16$ (frequency 4); $[25, 50)$ contains $25, 36, 49$ (3); $[50, 75)$: 64; $[75, 100)$: 81; $[100, 125)$: $100, 121$ (2); $[125, 150)$: 144; $[150, 175)$: 169; $[175, 200)$: 196; $[200, 225)$: none; $[225, 250)$: 225, because 225 is the left end point of this class.

The class marks are $12.5, 37.5, 62.5, \ldots, 237.5$.

$\sum fm = 4(12.5) + 3(37.5) + 62.5 + 87.5 + 2(112.5) + 137.5 + 162.5 + 187.5 + 237.5 = 1262.5$.

So $\bar{x}_g = \dfrac{1262.5}{15} = \dfrac{505}{6}$.

$\bar{x}_g - \bar{x} = \dfrac{505}{6} - \dfrac{496}{6} = \dfrac{9}{6} = \dfrac{3}{2}$.

Check by errors: $\sum(m_i - x_i) = 1262.5 - 1240 = 22.5$, and $\dfrac{22.5}{15} = \dfrac{3}{2}$. Placing 225 in $[200, 225)$ instead would give $-\dfrac{1}{6}$, which is option (B); the stated convention excludes it.

**Correct Option: (C)**

## Question 02

**English**

Let $a_1, a_2, a_3, \ldots$ be an arithmetic progression. The arithmetic mean of the ten observations $a_1, a_2, \ldots, a_{10}$ is 23, and the arithmetic mean of the twenty observations $a_1, a_2, \ldots, a_{20}$ is 43. Then the arithmetic mean of the twenty observations $a_{11}, a_{12}, \ldots, a_{30}$ is

(A) 83
(B) 63
(C) 73
(D) 93

Only one option is correct.

**हिन्दी**

माना $a_1, a_2, a_3, \ldots$ एक समान्तर श्रेढ़ी है। दस प्रेक्षणों $a_1, a_2, \ldots, a_{10}$ का समान्तर माध्य 23 है तथा बीस प्रेक्षणों $a_1, a_2, \ldots, a_{20}$ का समान्तर माध्य 43 है। तब बीस प्रेक्षणों $a_{11}, a_{12}, \ldots, a_{30}$ का समान्तर माध्य है

(A) 83
(B) 63
(C) 73
(D) 93

केवल एक विकल्प सही है।

**Solution**

Let the first term be $a$ and the common difference $d$.

The sum of the first $n$ terms is $S_n = \frac{n}{2}(2a + (n-1)d)$, so the mean of $a_1, \ldots, a_n$ is $\frac{S_n}{n} = a + \frac{(n-1)d}{2}$.

$n = 10$: $a + \frac{9d}{2} = 23$. $n = 20$: $a + \frac{19d}{2} = 43$.

Subtracting: $5d = 20$, so $d = 4$ and $a = 23 - 18 = 5$.

$a_{11} = 5 + 40 = 45$ and $a_{30} = 5 + 116 = 121$. The mean of the consecutive terms $a_{11}, \ldots, a_{30}$ is the mean of the first and last of them: $\frac{45 + 121}{2} = 83$.

Check: $S_{30} = 15(10 + 116) = 1890$, $S_{10} = 230$, and $\frac{1890 - 230}{20} = 83$. (The mean of $a_{11}, \ldots, a_{20}$ alone is 63, option (B).)

**Correct Option: (A)**

## Question 03

**English**

The marks of 30 students form the following continuous frequency distribution, in which each class contains its left end point but not its right end point.

| Class | $[0, 10)$ | $[10, 20)$ | $[20, 30)$ | $[30, 40)$ | $[40, 50)$ |
|---|---|---|---|---|---|
| Frequency | 4 | 6 | 9 | 7 | 4 |

Five more students obtain the marks $x$, $2x$, $x + 8$, 28 and 40, where $x$ is a real number, and the median of these five marks is 16. These five marks are entered in the same classes. Then the median of the resulting distribution of 35 students is

(A) 25
(B) 24.5
(C) 23.5
(D) 25.5

Only one option is correct.

**हिन्दी**

30 विद्यार्थियों के अंक निम्न सतत बारंबारता बंटन बनाते हैं, जिसमें प्रत्येक वर्ग में उसका बायाँ अन्त्य बिन्दु सम्मिलित है परन्तु दायाँ अन्त्य बिन्दु नहीं।

| Class | $[0,10)$ | $[10,20)$ | $[20,30)$ | $[30,40)$ | $[40,50)$ |
|---|---|---|---|---|---|
| Frequency | 4 | 6 | 9 | 7 | 4 |

पाँच अन्य विद्यार्थी अंक $x,\ 2x,\ x+8,\ 28$ तथा $40$ प्राप्त करते हैं, जहाँ $x$ एक वास्तविक संख्या है, तथा इन पाँच अंकों की माध्यिका $16$ है। इन पाँच अंकों को उन्हीं वर्गों में सम्मिलित किया जाता है। तब $35$ विद्यार्थियों के परिणामी बंटन की माध्यिका है

(A) 25
(B) 24.5
(C) 23.5
(D) 25.5

केवल एक विकल्प सही है।

**Solution**

The median of five observations is the third one in ascending order, so one of $x,\ 2x,\ x+8,\ 28,\ 40$ equals 16; it cannot be 28 or 40.

If $x = 16$: the marks are $16, 32, 24, 28, 40$, in order $16, 24, 28, 32, 40$, median $28 \neq 16$. Rejected.

If $2x = 16$, then $x = 8$; if $x + 8 = 16$, then also $x = 8$. The marks are $8, 16, 16, 28, 40$, whose median is 16. So $x = 8$.

Entering them: $8 \in [0,10)$, $16, 16 \in [10,20)$, $28 \in [20,30)$, $40 \in [40,50)$.

New frequencies $5, 8, 10, 7, 5$ with $N = 35$; cumulative frequencies $5, 13, 23, 30, 35$.

$\dfrac{N}{2} = 17.5$ lies in $[20,30)$, with $l = 20$, $C = 13$, $f = 10$, $h = 10$.

Median $= l + \dfrac{\frac{N}{2} - C}{f} h = 20 + \dfrac{4.5}{10} \cdot 10 = 24.5$.

(The median before the five students were added was $20 + \dfrac{15-10}{9} \cdot 10 \approx 25.56$.)

**Correct Option: (B)**

## Question 04

**English**

A continuous frequency distribution has the classes $[0,10), [10,20), [20,30), [30,40), [40,50)$. For $k = 0, 1, 2, 3, 4, 5$, the number of observations that are not less than $10k$ is $50 - 2k^2$. Then the median of this distribution is

(A) 32.5
(B) 36.25
(C) 37.5
(D) 35

Only one option is correct.

**हिन्दी**

एक सतत बारंबारता बंटन के वर्ग $[0, 10), [10, 20), [20, 30), [30, 40), [40, 50)$ हैं। $k = 0, 1, 2, 3, 4, 5$ के लिए, उन प्रेक्षणों की संख्या जो $10k$ से कम नहीं हैं, $50 - 2k^2$ है। तब इस बंटन की माध्यिका है

(A) 32.5
(B) 36.25
(C) 37.5
(D) 35

केवल एक विकल्प सही है।

**Solution**

The number of observations $\geq 10k$ for $k = 0, \ldots, 5$ is $50, 48, 42, 32, 18, 0$; so $N = 50$.

The frequency of $[10k, 10k + 10)$ is the number $\geq 10k$ minus the number $\geq 10(k + 1)$: $50 - 48 = 2$, $48 - 42 = 6$, $42 - 32 = 10$, $32 - 18 = 14$, $18 - 0 = 18$.

Less-than cumulative frequencies: $2, 8, 18, 32, 50$.

$\frac{N}{2} = 25$: the cumulative frequency first reaches 25 in $[30, 40)$, so $l = 30$, $C = 18$, $f = 14$, $h = 10$.

$$\text{Median} = 30 + \frac{25 - 18}{14} \cdot 10 = 30 + 5 = 35.$$

**Correct Option: (D)**

## Question 05

**English**

For each positive integer $n$, let $d(n)$ denote the number of positive divisors of $n$. Then the mode of the forty observations $d(1), d(2), d(3), \ldots, d(40)$ is

(A) 4
(B) 2
(C) 6
(D) 3

Only one option is correct.

**हिन्दी**

प्रत्येक धनात्मक पूर्णांक $n$ के लिए माना $d(n)$, $n$ के धनात्मक भाजकों की संख्या है। तब चालीस प्रेक्षणों $d(1), d(2), d(3), \ldots, d(40)$ का बहुलक है

(A) 4
(B) 2
(C) 6
(D) 3

केवल एक विकल्प सही है।

**Solution**

$d(n) = 2$ exactly when $n$ is prime. The primes up to 40 are $2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31, 37$: frequency 12.

$d(n) = 4$ exactly when $n = pq$ with distinct primes $p, q$, or $n = p^3$.

$n = pq \leq 40$: $6, 10, 14, 22, 26, 34, 38$ (with $p = 2$), $15, 21, 33, 39$ (with $p = 3$), $35$: twelve numbers.

$n = p^3$: $8, 27$. Frequency 14.

Remaining values: $d(1) = 1$; $d = 3$ for $4, 9, 25$; $d = 5$ for 16; $d = 6$ for $12, 18, 20, 28, 32$; $d = 8$ for $24, 30, 40$; $d = 9$ for 36.

Check: $12 + 14 + 1 + 3 + 1 + 5 + 3 + 1 = 40$.

The value 4 has the largest frequency ($14 > 12$), so the mode is 4, not 2.

**Correct Option: (A)**

## Question 06

**English**

A data set consists of the observation 6 occurring once, 12 occurring 4 times, 16 occurring $k$ times and 25 occurring 4 times, where $k$ is a positive integer. The same data are grouped into the classes $[0, 10), [10, 20), [20, 30)$. The ungrouped data have a unique mode, and this mode is equal to the mode of the grouped frequency distribution. Then $k$ is equal to

(A) 5
(B) 6
(C) 7
(D) 8

Only one option is correct.

**हिन्दी**

एक आँकड़ा समुच्चय में प्रेक्षण 6 एक बार, 12 चार बार, 16 $k$ बार तथा 25 चार बार आता है, जहाँ $k$ एक धनात्मक पूर्णांक है। इन्हीं आँकड़ों को वर्गों $[0, 10), [10, 20), [20, 30)$ में समूहित किया जाता है। अवर्गीकृत आँकड़ों का एक अद्वितीय बहुलक है तथा यह बहुलक समूहित बारंबारता बंटन के बहुलक के बराबर है। तब $k$ बराबर है

(A) 5
(B) 6
(C) 7
(D) 8

केवल एक विकल्प सही है।

**Solution**

Ungrouped data: 6 (once), 12 (4 times), 16 ($k$ times), 25 (4 times).

If $k \leq 4$, then 12 and 25 both occur 4 times and no value occurs more often, so the mode is not unique. Hence $k \geq 5$, and the unique mode is 16.

Grouped frequencies: $[0, 10)$: 1; $[10, 20)$: $4 + k$; $[20, 30)$: 4. As $4 + k \geq 9 > 4$, the modal class is $[10, 20)$ with $f_1 = 4 + k$, $f_0 = 1$, $f_2 = 4$, $h = 10$.

$$\text{Mode} = l + \frac{f_1 - f_0}{2f_1 - f_0 - f_2} h = 10 + \frac{(k + 3) \cdot 10}{2k + 3}.$$

Setting it equal to 16: $\frac{10(k + 3)}{2k + 3} = 6 \Rightarrow 10k + 30 = 12k + 18 \Rightarrow k = 6$.

Check: frequencies $1, 10, 4$ give $10 + \frac{9}{15} \cdot 10 = 16$, and 16 occurs 6 times, more than any other value.

**Correct Option: (B)**

## Question 07

**English**

In a continuous frequency distribution, the class $[10(k-1), 10k)$ has frequency $k^2(7-k)$ for $k = 1, 2, 3, 4, 5, 6$, and there are no other classes. Then the mode of this distribution is

(A) 40
(B) 42.5
(C) 45
(D) 41.25

Only one option is correct.

**हिन्दी**

एक सतत बारंबारता बंटन में $k = 1, 2, 3, 4, 5, 6$ के लिए वर्ग $[10(k-1), 10k)$ की बारंबारता $k^2(7-k)$ है, तथा अन्य कोई वर्ग नहीं है। तब इस बंटन का बहुलक है

(A) 40
(B) 42.5
(C) 45
(D) 41.25

केवल एक विकल्प सही है।

**Solution**

Frequencies $k^2(7-k)$: $k=1$: 6; $k=2$: 20; $k=3$: 36; $k=4$: 48; $k=5$: 50; $k=6$: 36.

The largest frequency is 50, for $k=5$, the class $[40, 50)$. So $l = 40$, $f_1 = 50$, $f_0 = 48$ (class $[30, 40)$), $f_2 = 36$ (class $[50, 60)$), $h = 10$.

$$\text{Mode} = l + \frac{f_1 - f_0}{2f_1 - f_0 - f_2} h = 40 + \frac{2}{16} \cdot 10 = 41.25.$$

The mode is close to 40 because the neighbouring class on the left has almost the same frequency.

**Correct Option: (D)**

## Question 08

**English**

Consider the ten observations $1, 2, 3, 4, 5, 6, 7, 8, 9, 40$. Let $D_1$ be their mean deviation about the mean and $D_2$ their mean deviation about the median. Then $D_1 - D_2$ is equal to

(A) 0.5
(B) 1.4
(C) 0.9
(D) 0

Only one option is correct.

**हिन्दी**

दस प्रेक्षणों $1, 2, 3, 4, 5, 6, 7, 8, 9, 40$ पर विचार कीजिए। माना $D_1$ इनका माध्य के सापेक्ष माध्य विचलन है तथा $D_2$ इनका माध्यिका के सापेक्ष माध्य विचलन है। तब $D_1 - D_2$ बराबर है

(A) 0.5
(B) 1.4
(C) 0.9
(D) 0

केवल एक विकल्प सही है।

**Solution**

Mean $\bar{x} = \dfrac{1 + 2 + \cdots + 9 + 40}{10} = \dfrac{85}{10} = 8.5$. In order, the 5th and 6th observations are 5 and 6, so the median is $M = 5.5$.

About the mean: $|1 - 8.5| + \cdots + |8 - 8.5| = 7.5 + 6.5 + 5.5 + 4.5 + 3.5 + 2.5 + 1.5 + 0.5 = 32$, $|9 - 8.5| = 0.5$, $|40 - 8.5| = 31.5$. Total 64, so $D_1 = 6.4$.

About the median: $|1 - 5.5|, \ldots, |9 - 5.5|$ are $4.5, 3.5, 2.5, 1.5, 0.5, 0.5, 1.5, 2.5, 3.5$, total 20.5; $|40 - 5.5| = 34.5$. Total 55, so $D_2 = 5.5$.

$D_1 - D_2 = 6.4 - 5.5 = 0.9$. As expected, the mean deviation is smallest about the median, and the outlier 40 pulls the mean away from the bulk of the data.

**Correct Option: (C)**

## Question 09

**English**

Let $x$ be a real number such that the mean deviation about the median of the five observations $3, 7, 10, 14, x$ is 4. Then the sum of all possible values of $x$ is

(A) 17
(B) 16
(C) 1
(D) 18

Only one option is correct.

**हिन्दी**

माना $x$ एक वास्तविक संख्या है जिसके लिए पाँच प्रेक्षणों $3, 7, 10, 14, x$ का माध्यिका के सापेक्ष माध्य विचलन $4$ है। तब $x$ के सभी सम्भव मानों का योग है

(A) 17
(B) 16
(C) 1
(D) 18

केवल एक विकल्प सही है।

**Solution**

The mean deviation about the median $M$ is $\frac{1}{5}\sum |x_i - M| = 4$, so $\sum |x_i - M| = 20$.

The median is the third value in order, so it depends on where $x$ lies.

Case $x \leq 7$: the order puts 7 third, $M = 7$. Sum $= |3-7| + 0 + |10-7| + |14-7| + |x-7| = 14 + 7 - x = 21 - x$. Setting $21 - x = 20$ gives $x = 1 \leq 7$. Valid.

Case $7 \leq x \leq 10$: $M = x$. Sum $= (x-3) + (x-7) + (10-x) + (14-x) = 14$, a constant, never 20.

Case $x \geq 10$: $M = 10$. Sum $= 7 + 3 + 0 + 4 + (x-10) = x + 4$. Setting $x + 4 = 20$ gives $x = 16 \geq 10$. Valid.

So $x \in \{1, 16\}$, and the sum is 17.

**Correct Option: (A)**

## Question 10

**English**

Group $P$ has 40 observations with mean 50 and variance 9. Group $Q$ has 60 observations with variance 16, and the mean of $Q$ is greater than the mean of $P$. The variance of the 100 observations of $P$ and $Q$ taken together is 19.2. Then the mean of the observations of $Q$ is

(A) 45
(B) 52.5
(C) 55
(D) 60

Only one option is correct.

**हिन्दी**

समूह $P$ में 40 प्रेक्षण हैं जिनका माध्य 50 तथा प्रसरण 9 है। समूह $Q$ में 60 प्रेक्षण हैं जिनका प्रसरण 16 है, तथा $Q$ का माध्य $P$ के माध्य से अधिक है। $P$ तथा $Q$ के सभी 100 प्रेक्षणों को साथ लेने पर प्रसरण 19.2 है। तब $Q$ के प्रेक्षणों का माध्य है

(A) 45
(B) 52.5
(C) 55
(D) 60

केवल एक विकल्प सही है।

**Solution**

Let $\bar{x}_Q = 50 + d$. The combined mean is $\bar{x} = \dfrac{40 \cdot 50 + 60(50 + d)}{100} = 50 + 0.6d$.

Deviations of the group means from $\bar{x}$: $d_P = -0.6d$, $d_Q = 0.4d$.

Combined variance $= \dfrac{n_P(\sigma_P^2 + d_P^2) + n_Q(\sigma_Q^2 + d_Q^2)}{n_P + n_Q} = \dfrac{40(9 + 0.36d^2) + 60(16 + 0.16d^2)}{100} = 13.2 + 0.24d^2$.

$13.2 + 0.24d^2 = 19.2 \Rightarrow d^2 = 25 \Rightarrow d = \pm 5$.

Since the mean of $Q$ exceeds that of $P$, $d = 5$ and $\bar{x}_Q = 55$. (Without that condition, 45 would also fit.)

**Correct Option: (C)**

## Question 11

**English**

Two groups of observations have the same coefficient of variation 20%. The first group has 30 observations with mean 40, and the second group has 20 observations with mean 60. Then the coefficient of variation, in per cent, of all 50 observations taken together is

(A) $\frac{25\sqrt{6}}{3}$
(B) $\frac{50}{\sqrt{3}}$
(C) 20
(D) $\frac{100}{3}$

Only one option is correct.

**हिन्दी**

प्रेक्षणों के दो समूहों का विचरण गुणांक समान, $20\%$, है। पहले समूह में $30$ प्रेक्षण हैं जिनका माध्य $40$ है, तथा दूसरे समूह में $20$ प्रेक्षण हैं जिनका माध्य $60$ है। तब सभी $50$ प्रेक्षणों को साथ लेने पर विचरण गुणांक (प्रतिशत में) है

(A) $\frac{25\sqrt{6}}{3}$
(B) $\frac{50}{\sqrt{3}}$
(C) 20
(D) $\frac{100}{3}$

केवल एक विकल्प सही है।

**Solution**

CV $= \frac{\sigma}{\bar{x}} \times 100$. With CV $= 20$: $\sigma_1 = 8$ and $\sigma_2 = 12$.

Combined mean: $\bar{x} = \frac{30 \cdot 40 + 20 \cdot 60}{50} = 48$.

Combined variance: $\sigma^2 = \frac{n_1(\sigma_1^2 + (\bar{x}_1 - \bar{x})^2) + n_2(\sigma_2^2 + (\bar{x}_2 - \bar{x})^2)}{n_1 + n_2} = \frac{30(64 + 64) + 20(144 + 144)}{50} = \frac{3840 + 5760}{50} = 192$.

So $\sigma = \sqrt{192} = 8\sqrt{3}$.

Combined CV $= \frac{8\sqrt{3}}{48} \times 100 = \frac{50}{\sqrt{3}} \approx 28.87$.

So pooling two equally consistent groups with different means gives a less consistent whole; the within-group part alone gives $\sqrt{96}/48 \times 100 = \frac{25\sqrt{6}}{3}$, and averaging the CVs gives 20, both wrong.

**Correct Option: (B)**

## Question 12

**English**

Consider the $n$ observations $3, 7, 11, \ldots, 4n-1$, which are in arithmetic progression. When one more observation, equal to the arithmetic mean of these $n$ observations, is included, the variance of the $n+1$ observations is 120. Then the largest of the original $n$ observations is

(A) 35
(B) 43
(C) 41
(D) 39

Only one option is correct.

**हिन्दी**

समान्तर श्रेढ़ी में स्थित $n$ प्रेक्षणों $3, 7, 11, \ldots, 4n-1$ पर विचार कीजिए। जब इन $n$ प्रेक्षणों के समान्तर माध्य के बराबर एक और प्रेक्षण सम्मिलित किया जाता है, तब $n+1$ प्रेक्षणों का प्रसरण 120 है। तब मूल $n$ प्रेक्षणों में सबसे बड़ा प्रेक्षण है

(A) 35
(B) 43
(C) 41
(D) 39

केवल एक विकल्प सही है।

**Solution**

For $n$ terms of an AP with common difference $d$, the variance is $\dfrac{d^2(n^2-1)}{12}$; here $d=4$, so $\sigma^2 = \dfrac{4(n^2-1)}{3}$.

Hence $\sum_{i=1}^{n}(x_i-\bar{x})^2 = n\sigma^2 = \dfrac{4n(n^2-1)}{3}$.

The extra observation equals $\bar{x}$, so the mean of the $n+1$ observations is still $\bar{x}$, and the extra deviation is 0.

New variance $= \dfrac{1}{n+1}\cdot\dfrac{4n(n^2-1)}{3} = \dfrac{4n(n-1)}{3}$.

$\dfrac{4n(n-1)}{3} = 120 \Rightarrow n^2 - n - 90 = 0 \Rightarrow (n-10)(n+9) = 0 \Rightarrow n = 10.$

Check: the AP $3, 7, \ldots, 39$ has mean 21 and variance 132; with 21 added the variance is $\dfrac{10\cdot 132}{11} = 120$.

The largest original observation is $4\cdot 10 - 1 = 39$.

**Correct Option: (D)**

## Question 13

**English**

Consider the fifteen observations $k^2$, $k = -7, -6, \ldots, 6, 7$. For these observations, let $M$ be the median, and let the first and third quartiles $Q_1$ and $Q_3$ be the 4th and the 12th observations, respectively, when the observations are arranged in ascending order. If $Q$ denotes the quartile deviation, then $\dfrac{Q}{M}$ is equal to

(A) $\dfrac{1}{2}$
(B) 1
(C) $\dfrac{5}{3}$
(D) 2

Only one option is correct.

**हिन्दी**

पंद्रह प्रेक्षणों $k^2$, $k = -7, -6, \ldots, 6, 7$ पर विचार कीजिए। इन प्रेक्षणों के लिए माना $M$ माध्यिका है, तथा प्रेक्षणों को आरोही क्रम में व्यवस्थित करने पर प्रथम तथा तृतीय चतुर्थक $Q_1$ तथा $Q_3$ क्रमशः 4वाँ तथा 12वाँ प्रेक्षण हैं। यदि $Q$ चतुर्थक विचलन को दर्शाता है, तो $\dfrac{Q}{M}$ बराबर है

(A) $\dfrac{1}{2}$
(B) 1
(C) $\dfrac{5}{3}$
(D) 2

केवल एक विकल्प सही है।

**Solution**

The values are 0 once and $1, 4, 9, 16, 25, 36, 49$ twice each. Ascending order: $0, 1, 1, 4, 4, 9, 9, 16, 16, 25, 25, 36, 36, 49,$

With 15 observations the median is the 8th: $M = 16$.

$Q_1$ is the 4th observation: 4. $Q_3$ is the 12th observation: 36.

Quartile deviation $Q = \dfrac{Q_3 - Q_1}{2} = \dfrac{36 - 4}{2} = 16.$

So $\dfrac{Q}{M} = 1$. (The ratio $\dfrac{Q_3 - M}{M - Q_1} = \dfrac{20}{12} = \dfrac{5}{3}$ is option (C), a common confusion.)

**Correct Option: (B)**

## Question 14

**English**

Six observations $a, ar, ar^2, ar^3, ar^4, ar^5$, where $a > 0$ and $r$ is a positive real number with $r \neq 1$, are in geometric progression. The arithmetic mean of all six observations is $\frac{4}{5}$ times the arithmetic mean of the three observations $a, ar^2, ar^4$. Then the value of $r$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

**हिन्दी**

छः प्रेक्षण $a, ar, ar^2, ar^3, ar^4, ar^5$, जहाँ $a > 0$ तथा $r$ एक धनात्मक वास्तविक संख्या है और $r \neq 1$, गुणोत्तर श्रेढ़ी में हैं। सभी छः प्रेक्षणों का समान्तर माध्य, तीन प्रेक्षणों $a, ar^2, ar^4$ के समान्तर माध्य का $\frac{4}{5}$ गुना है। तब $r$ का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

**Solution**

Sum of the GP: $S_6 = a\dfrac{r^6 - 1}{r - 1}$ for $r \neq 1$.

Factorise: $r^6 - 1 = (r^2 - 1)(r^4 + r^2 + 1) = (r - 1)(r + 1)(r^4 + r^2 + 1)$, so $S_6 = a(1 + r)(1 + r^2 + r^4)$.

Mean of all six: $\dfrac{a(1 + r)(1 + r^2 + r^4)}{6}$. Mean of $a, ar^2, ar^4$: $\dfrac{a(1 + r^2 + r^4)}{3}$.

Condition: $\dfrac{a(1 + r)(1 + r^2 + r^4)}{6} = \dfrac{4}{5} \cdot \dfrac{a(1 + r^2 + r^4)}{3}$.

Divide by $a(1 + r^2 + r^4) > 0$: $\dfrac{1 + r}{6} = \dfrac{4}{15}$, so $1 + r = \dfrac{8}{5}$ and $r = \dfrac{3}{5}$.

Check with $a = 1$: the six terms sum to $\dfrac{1 - 0.6^6}{0.4} = 2.38336$, mean $0.397227$; the odd-position terms $1, 0.36, 0.1296$ have mean $0.496533$, and $0.8 \times 0.496533 = 0.397227$.

**Final Answer: 0.6**

## Question 15

**English**

The marks of the students of section $A$ form the following continuous frequency distribution, in which each class contains its left end point but not its right end point, and $f$ is a positive integer.

| Class | $[0, 20)$ | $[20, 40)$ | $[40, 60)$ | $[60, 80)$ |
|---|---|---|---|---|
| Frequency | 3 | 7 | $f$ | 5 |

Section $B$ has 25 students whose mean mark is 52. The arithmetic mean of the marks of all the students of the two sections taken together, computed with the class marks for section $A$, is 48. Then the number of students of section $A$ who scored at least 40 marks is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

**हिन्दी**

अनुभाग $A$ के विद्यार्थियों के अंक निम्न सतत बारंबारता बंटन बनाते हैं, जिसमें प्रत्येक वर्ग में उसका बायाँ अन्त्य बिन्दु सम्मिलित है परन्तु दायाँ अन्त्य बिन्दु नहीं, तथा $f$ एक धनात्मक पूर्णांक है।

| Class | $[0, 20)$ | $[20, 40)$ | $[40, 60)$ | $[60, 80)$ |
|---|---|---|---|---|
| Frequency | 3 | 7 | $f$ | 5 |

अनुभाग $B$ में 25 विद्यार्थी हैं जिनके अंकों का माध्य 52 है। दोनों अनुभागों के सभी विद्यार्थियों के अंकों का समान्तर माध्य,

अनुभाग $A$ के लिए वर्ग चिह्नों से परिकलित, $48$ है। तब अनुभाग $A$ के उन विद्यार्थियों की संख्या जिन्होंने कम से कम $40$ अंक प्राप्त किए, _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

**Solution**

Class marks of section $A$: $10, 30, 50, 70$. Total of section $A$: $30 + 210 + 50f + 350 = 590 + 50f$, with $15 + f$ students.

Total of section $B$: $25 \times 52 = 1300$.

Combined mean $= \dfrac{\text{total of } A + \text{total of } B}{n_A + n_B} = \dfrac{1890 + 50f}{40 + f} = 48.$

$1890 + 50f = 1920 + 48f \Rightarrow 2f = 30 \Rightarrow f = 15.$

Check: section $A$ has 30 students with mean $\dfrac{1340}{30}$; $\dfrac{1340 + 1300}{55} = 48.$

Students of $A$ with at least 40 marks are those in $[40, 60)$ and $[60, 80)$: $15 + 5 = 20$.

**Final Answer: 20**

## Question 16

**English**

A continuous frequency distribution has the following classes and frequencies.

| Class | $[0, 10)$ | $[10, 20)$ | $[20, 30)$ | $[30, 40)$ | $[40, 50)$ |
|---|---|---|---|---|---|
| Frequency | 1 | 4 | 6 | 4 | 1 |

Let $k$ be a real number with $k < 10$. The variance of the five observations $10, 20, 30, 40, k$ is five times the variance of the above distribution, computed with the class marks. Then the value of $k$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

**हिन्दी**

एक सतत बारंबारता बंटन के वर्ग तथा बारंबारताएँ निम्न हैं।

| Class | $[0, 10)$ | $[10, 20)$ | $[20, 30)$ | $[30, 40)$ | $[40, 50)$ |
|---|---|---|---|---|---|
| Frequency | 1 | 4 | 6 | 4 | 1 |

माना $k$ एक वास्तविक संख्या है जहाँ $k < 10$ है। पाँच प्रेक्षणों $10, 20, 30, 40, k$ का प्रसरण, वर्ग चिह्नों से परिकलित उपरोक्त बंटन के प्रसरण का पाँच गुना है। तब $k$ का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

**Solution**

For the distribution, the class marks are $5, 15, 25, 35, 45$. With $u = \dfrac{m - 25}{10}$: $u = -2, -1, 0, 1, 2$, frequencies $1, 4, 6, 4, 1$, $N = 16$.

$\sum fu = 0$ and $\sum fu^2 = 4 + 4 + 0 + 4 + 4 = 16$, so $\sigma_u^2 = 1$ and $\sigma^2 = 10^2 \cdot 1 = 100$.

For $10, 20, 30, 40, k$: $\sum x = 100 + k$, $\sum x^2 = 3000 + k^2$.

$$\sigma^2 = \frac{3000 + k^2}{5} - \frac{(100 + k)^2}{25} = \frac{15000 + 5k^2 - 10000 - 200k - k^2}{25} = \frac{4k^2 - 200k + 5000}{25}.$$

Setting this equal to $5 \times 100 = 500$: $4k^2 - 200k + 5000 = 12500 \Rightarrow k^2 - 50k - 1875 = 0$.

$k = \dfrac{50 \pm \sqrt{2500 + 7500}}{2} = \dfrac{50 \pm 100}{2}$, so $k = 75$ or $k = -25$.

Since $k < 10$, $k = -25$. Check: $10, 20, 30, 40, -25$ have mean 15 and $\sum(x-15)^2 = 25+25+225+625+1600 = 2500$, variance 500.

**Final Answer: -25**

## Question 17

**English**

For a real number $t > 0$, let $V(t)$ denote the variance of the four observations $t,\ 2t,\ \dfrac{576}{t},\ \dfrac{1152}{t}$. Then the minimum value of $V(t)$ over all $t > 0$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

**हिन्दी**

वास्तविक संख्या $t > 0$ के लिए माना $V(t)$ चार प्रेक्षणों $t,\ 2t,\ \dfrac{576}{t},\ \dfrac{1152}{t}$ का प्रसरण है। तब सभी $t > 0$ पर $V(t)$ का न्यूनतम मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

**Solution**

The observations are $t,\ \dfrac{576}{t}$ and twice each of these. Let $s = t + \dfrac{576}{t} > 0$.

Sum $= 3t + \dfrac{3 \cdot 576}{t} = 3s$, so the mean is $\dfrac{3s}{4}$.

Sum of squares $= t^2 + 4t^2 + \dfrac{576^2}{t^2} + \dfrac{4 \cdot 576^2}{t^2} = 5\left(t^2 + \dfrac{576^2}{t^2}\right) = 5(s^2 - 2 \cdot 576)$.

$V = \dfrac{5(s^2 - 1152)}{4} - \dfrac{9s^2}{16} = \dfrac{20s^2 - 23040 - 9s^2}{16} = \dfrac{11s^2 - 23040}{16}$.

$V$ increases with $s$ for $s > 0$, so $V$ is least when $s$ is least.

By AM-GM, $t + \dfrac{576}{t} \geq 2\sqrt{576} = 48$, with equality when $t = \dfrac{576}{t}$, i.e. $t = 24$.

$V_{\min} = \dfrac{11 \cdot 48^2 - 23040}{16} = \dfrac{25344 - 23040}{16} = 144$.

Check: at $t = 24$ the observations are $24, 48, 24, 48$, mean 36, variance $\dfrac{4 \cdot 144}{4} = 144$.

**Final Answer: 144**

## Question 18

**English**

Let $x_1, x_2, x_3, x_4, x_5$ be the observations $-3, 1, 3, 4, 5$, respectively. For a real number $\lambda$, let $V(\lambda)$ denote the variance of the five observations $y_i = x_i^2 + \lambda x_i$, $i = 1, 2, 3, 4, 5$. Then the value of $\lambda$ for which $V(\lambda)$ is minimum is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

**हिन्दी**

माना $x_1, x_2, x_3, x_4, x_5$ क्रमशः प्रेक्षण $-3, 1, 3, 4, 5$ हैं। वास्तविक संख्या $\lambda$ के लिए माना $V(\lambda)$ पाँच प्रेक्षणों $y_i = x_i^2 + \lambda x_i$, $i = 1, 2, 3, 4, 5$ का प्रसरण है। तब $\lambda$ का वह मान जिसके लिए $V(\lambda)$ न्यूनतम है, _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

**Solution**

$\bar{y} = \overline{x^2} + \lambda\bar{x}$, so $y_i - \bar{y} = (x_i^2 - \overline{x^2}) + \lambda(x_i - \bar{x})$.

Squaring and averaging: $V(\lambda) = A + 2B\lambda + C\lambda^2$, where $A = \frac{1}{5}\sum(x_i^2 - \overline{x^2})^2$, $B = \frac{1}{5}\sum(x_i^2 - \overline{x^2})(x_i - \bar{x})$, $C = \frac{1}{5}\sum(x_i - \bar{x})^2$.

Data $-3, 1, 3, 4, 5$: $\sum x = 10$, $\sum x^2 = 9+1+9+16+25 = 60$, $\sum x^3 = -27+1+27+64+125 = 190$.

$\bar{x} = 2$, $\overline{x^2} = 12$, $\overline{x^3} = 38$. $C = \overline{x^2} - \bar{x}^2 = 8$ (the variance of the $x_i$).

$B = \overline{x^3} - \overline{x^2}\,\bar{x} = 38 - 24 = 14$ (expanding the product and using $\sum(x_i - \bar{x}) = 0$).

$V(\lambda) = C\lambda^2 + 2B\lambda + A$ is a quadratic with positive leading coefficient, so its minimum over $\mathbb{R}$ is at $\lambda = -\dfrac{2B}{2C} = -\dfrac{14}{8} = -1.75$.

Check directly: $V(\lambda) = 8\lambda^2 + 28\lambda + \dfrac{324}{5}$, and $V'(\lambda) = 16\lambda + 28 = 0$ gives $\lambda = -1.75$.

**Final Answer: -1.75**

## Question 19

**English**

The median of the eight observations $9, 11, 13, 14, 27, 31, 37, x$, where $x$ is a real number, is 16. Then the mean deviation about the mean of these eight observations is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

**हिन्दी**

आठ प्रेक्षणों $9, 11, 13, 14, 27, 31, 37, x$, जहाँ $x$ एक वास्तविक संख्या है, की माध्यिका $16$ है। तब इन आठ प्रेक्षणों का माध्य के सापेक्ष माध्य विचलन _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

**Solution**

With eight observations, the median is the mean of the 4th and 5th in ascending order.

If $x \le 13$: 4th and 5th are $13, 14$, median $13.5 \ne 16$. If $13 \le x \le 14$: 4th and 5th are $x, 14$, median at most 14. If $x \ge 27$: 4th and 5th are $14, 27$, median $20.5 \ne 16$.

So $14 \le x \le 27$, the 4th and 5th are 14 and $x$, and $\dfrac{14 + x}{2} = 16$ gives $x = 18$.

Mean $= \dfrac{160}{8} = 20$.

Absolute deviations from 20: $11, 9, 7, 6, 2, 7, 11, 17$; total 70.

Mean deviation about the mean $= \dfrac{70}{8} = 8.75$.

**Final Answer: 8.75**

## Question 20

**English**

Let $f : \mathbb{R} \to \mathbb{R}$ be defined by $f(x) = x^3 - 11x$. For $t \in [-3, 0]$, let $D(t)$ denote the mean deviation about the mean of the four observations $t,\ -t,\ f(t),\ f(-t)$. Then the value of $t \in [-3, 0]$ at which $D(t)$ attains its maximum value is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

**हिन्दी**

माना $f : \mathbb{R} \to \mathbb{R}$, $f(x) = x^3 - 11x$ द्वारा परिभाषित है। $t \in [-3, 0]$ के लिए माना $D(t)$ चार प्रेक्षणों $t,\ -t,\ f(t),\ f(-t)$ का माध्य के सापेक्ष माध्य विचलन है। तब $t \in [-3, 0]$ का वह मान जिस पर $D(t)$ अपना अधिकतम मान प्राप्त करता है, _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

**Solution**

$f(-x) = -x^3 + 11x = -f(x)$, so $f(-t) = -f(t)$ and the observations are $t, -t, f(t), -f(t)$, with sum 0 and mean 0.

$D(t) = \dfrac{|t| + |-t| + |f(t)| + |-f(t)|}{4} = \dfrac{|t| + |f(t)|}{2}$.

For $t \in [-3, 0]$: $|t| = -t$. Also $f(t) = t(t^2 - 11)$ with $t \le 0$ and $t^2 - 11 \le -2 < 0$, so $f(t) \ge 0$ and $|f(t)| = t^3 - 11t$.

Hence $D(t) = \dfrac{-t + t^3 - 11t}{2} = \dfrac{t^3 - 12t}{2}$ on $[-3, 0]$.

$D'(t) = \dfrac{3t^2 - 12}{2}$, which is zero at $t = -2$ in $[-3, 0]$. $D'$ is positive on $(-3, -2)$ and negative on

$(-2, 0)$ (first derivative test); also $D''(t) = 3t$, $D''(-2) = -6 < 0$.
$D(-2) = \dfrac{-8+24}{2} = 8$, $D(-3) = \dfrac{-27+36}{2} = 4.5$, $D(0) = 0$.
So the maximum of $D$ on $[-3, 0]$ is 8, attained at $t = -2$.

**Final Answer: -2**

## Question 21

**English**

The following continuous frequency distribution, in which $f$ is a positive integer, has arithmetic mean 28.5 (computed with the class marks).

| Class | $[0, 10)$ | $[10, 20)$ | $[20, 30)$ | $[30, 40)$ | $[40, 50)$ |
|---|---|---|---|---|---|
| Frequency | 2 | 3 | 5 | $f$ | 4 |

Then the mean deviation about the mean of this distribution is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

**हिन्दी**

निम्न सतत बारंबारता बंटन, जिसमें $f$ एक धनात्मक पूर्णांक है, का समान्तर माध्य (वर्ग चिह्नों से परिकलित) 28.5 है।

| Class | $[0, 10)$ | $[10, 20)$ | $[20, 30)$ | $[30, 40)$ | $[40, 50)$ |
|---|---|---|---|---|---|
| Frequency | 2 | 3 | 5 | $f$ | 4 |

तब इस बंटन का माध्य के सापेक्ष माध्य विचलन _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

**Solution**

Class marks: $5, 15, 25, 35, 45$. $\sum fm = 10 + 45 + 125 + 35f + 180 = 360 + 35f$ and $N = 14 + f$.
$\dfrac{360+35f}{14+f} = 28.5 \Rightarrow 360 + 35f = 399 + 28.5f \Rightarrow 6.5f = 39 \Rightarrow f = 6$, so $N = 20$.
Absolute deviations of the class marks from 28.5: $23.5, 13.5, 3.5, 6.5, 16.5$.
$\sum f|m - 28.5| = 2(23.5) + 3(13.5) + 5(3.5) + 6(6.5) + 4(16.5) = 47 + 40.5 + 17.5 + 39 + 66 = 210$.
Mean deviation about the mean $= \dfrac{210}{20} = 10.5$.

**Final Answer: 10.5**

## Question 22

**English**

The following continuous frequency distribution, in which $k$ is a positive integer, has arithmetic mean 32.5 (computed with the class marks).

| Class | $[0, 10)$ | $[10, 20)$ | $[20, 30)$ | $[30, 40)$ | $[40, 50)$ |
|---|---|---|---|---|---|
| Frequency | 1 | 2 | 3 | 4 | $k$ |

Let $D$ be the mean deviation about the mean and $\sigma$ the standard deviation of this distribution. Then the value of $\dfrac{D}{\sigma}$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

**हिन्दी**

निम्न सतत बारंबारता बंटन, जिसमें $k$ एक धनात्मक पूर्णांक है, का समान्तर माध्य (वर्ग चिह्नों से परिकलित) $32.5$ है।

| Class | $[0,10)$ | $[10,20)$ | $[20,30)$ | $[30,40)$ | $[40,50)$ |
|---|---|---|---|---|---|
| Frequency | 1 | 2 | 3 | 4 | $k$ |

माना $D$ इस बंटन का माध्य के सापेक्ष माध्य विचलन है तथा $\sigma$ इसका मानक विचलन है। तब $\dfrac{D}{\sigma}$ का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

**Solution**

Class marks $5, 15, 25, 35, 45$; take $u = \dfrac{m-5}{10} = 0, 1, 2, 3, 4$.

Mean: $\bar{u} = \dfrac{0+2+6+12+4k}{10+k} = \dfrac{32.5-5}{10} = 2.75$, so $20 + 4k = 27.5 + 2.75k$, $1.25k = 7.5$, $k = 6$ and $N = 16$.

Mean deviation: $|m - 32.5| = 27.5, 17.5, 7.5, 2.5, 12.5$; $\sum f|m-32.5| = 1(27.5) + 2(17.5) + 3(7.5) + 4(2.5) + 6(12.5) = 170$; $D = \dfrac{170}{16} = 10.625$.

Variance: $\sum fu^2 = 0 + 2 + 12 + 36 + 96 = 146$, $\sigma_u^2 = \dfrac{146}{16} - \left(\dfrac{11}{4}\right)^2 = \dfrac{146-121}{16} = \dfrac{25}{16}$.

$\sigma = 10\sigma_u = 10 \cdot \dfrac{5}{4} = 12.5$ (variance $156.25$).

$\dfrac{D}{\sigma} = \dfrac{10.625}{12.5} = 0.85$, less than 1 as it must be.

**Final Answer: 0.85**

## Question 23

**English**

The observations $x_1, x_2, x_3, x_4, x_5$ are $3, 7, 8, 12, 15$, respectively. Let $a$ and $b$ be real numbers with $a < 0$, and let $y_i = ax_i + b$ for $i = 1, 2, 3, 4, 5$. If the mean of $y_1, y_2, \ldots, y_5$ is 2 and their variance is 4.3, then the value of $ab$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

**हिन्दी**

प्रेक्षण $x_1, x_2, x_3, x_4, x_5$ क्रमशः $3, 7, 8, 12, 15$ हैं। माना $a$ तथा $b$ वास्तविक संख्याएँ हैं जहाँ $a < 0$ है, तथा $i = 1, 2, 3, 4, 5$ के लिए $y_i = ax_i + b$ है। यदि $y_1, y_2, \ldots, y_5$ का माध्य 2 है तथा इनका प्रसरण $4.3$ है, तो $ab$ का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

**Solution**

$\bar{x} = \dfrac{3+7+8+12+15}{5} = 9$. Deviations from 9: $-6, -2, -1, 3, 6$; squares $36, 4, 1, 9, 36$, sum 86.

$\sigma_x^2 = \dfrac{86}{5} = 17.2$.

Under $y = ax + b$ the mean becomes $a\bar{x} + b$ and the variance becomes $a^2\sigma_x^2$ (the shift $b$ does not change the variance; the scale $a$ multiplies it by $a^2$).

$a^2 \cdot 17.2 = 4.3 \Rightarrow a^2 = \dfrac{1}{4}$. As $a < 0$, $a = -\dfrac{1}{2}$.

$9a + b = 2 \Rightarrow b = 2 + \dfrac{9}{2} = \dfrac{13}{2}$.

$ab = -\frac{1}{2} \cdot \frac{13}{2} = -\frac{13}{4} = -3.25.$

**Final Answer: -3.25**

## Question 24

**English**

For ten observations $x_1, x_2, \ldots, x_{10}$, the values $u_i = \frac{x_i - 50}{2}$ were computed, and the mean and the variance of $u_1, u_2, \ldots, u_{10}$ were found to be $-3$ and 4, respectively. Later it was found that one observation had been recorded as 40 instead of its correct value 60. Then the correct variance of $x_1, x_2, \ldots, x_{10}$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

**हिन्दी**

दस प्रेक्षणों $x_1, x_2, \ldots, x_{10}$ के लिए मान $u_i = \frac{x_i - 50}{2}$ परिकलित किए गए, तथा $u_1, u_2, \ldots, u_{10}$ का माध्य तथा प्रसरण क्रमशः $-3$ तथा $4$ पाए गए। बाद में ज्ञात हुआ कि एक प्रेक्षण अपने सही मान $60$ के स्थान पर $40$ अंकित किया गया था। तब $x_1, x_2, \ldots, x_{10}$ का सही प्रसरण _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

**Solution**

From the recorded values: $\sum u_i = 10(-3) = -30$ and $\sum u_i^2 = 10\left(\sigma_u^2 + \bar{u}^2\right) = 10(4 + 9) = 130$.

The wrong observation 40 gave $u = \frac{40 - 50}{2} = -5$; the correct value 60 gives $u = 5$.

Corrected sums: $\sum u = -30 - (-5) + 5 = -20$, $\sum u^2 = 130 - 25 + 25 = 130$ (the square does not change).

Corrected mean of $u$: $-2$; corrected variance of $u$: $\frac{130}{10} - (-2)^2 = 9$.

Since $x_i = 2u_i + 50$, the variance of $x$ is $2^2 \cdot 9 = 36$ (the origin 50 does not affect it).

(The correct mean of $x$ is $50 + 2(-2) = 46$, and the recorded variance of $x$ was 16.)

**Final Answer: 36**

## Question 25

**English**

The arithmetic mean of 40 observations was computed as 152. It was later found that one of the observations, a three-digit number with no zero digit, had been recorded with its digits in the reverse order, and that the correct arithmetic mean is 156.95. The recorded value has hundreds digit 3 and tens digit 7. Then the correct value of this observation is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

**हिन्दी**

40 प्रेक्षणों का समान्तर माध्य 152 परिकलित किया गया। बाद में ज्ञात हुआ कि प्रेक्षणों में से एक, जो एक तीन अंकों की संख्या है जिसमें कोई अंक शून्य नहीं है, अपने अंकों के उल्टे क्रम में अंकित किया गया था, तथा सही समान्तर माध्य 156.95 है। अंकित मान का सैकड़े का अंक 3 तथा दहाई का अंक 7 है। तब इस प्रेक्षण का सही मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

**Solution**

Recorded total $= 40 \times 152 = 6080$; correct total $= 40 \times 156.95 = 6278$. The correct observation exceeds the recorded one by 198.

The recorded value is $\overline{37c} = 370 + c$ with $1 \le c \le 9$; reversing the digits gives the correct value $\overline{c73} = 100c + 73$.

$(100c + 73) - (370 + c) = 99c - 297 = 198 \Rightarrow 99c = 495 \Rightarrow c = 5$.

The recorded value was 375 and the correct value is 573; check: $573 - 375 = 198$.

**Final Answer: 573**

## Question 26

**English**

The four observations $13, 19, 21, x$, where $x$ is an integer, have coefficient of variation 25%. When the same real number $c$ is added to each of these four observations, the coefficient of variation of the new observations becomes 40%. Then the value of $c$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

**हिन्दी**

चार प्रेक्षणों $13, 19, 21, x$, जहाँ $x$ एक पूर्णांक है, का विचरण गुणांक 25% है। जब इन चारों प्रेक्षणों में से प्रत्येक में एक ही वास्तविक संख्या $c$ जोड़ी जाती है, तब नए प्रेक्षणों का विचरण गुणांक 40% हो जाता है। तब $c$ का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

**Solution**

$\text{CV} = \dfrac{\sigma}{\bar{x}} \times 100 = 25$ means $\sigma = \dfrac{\bar{x}}{4}$, i.e. $16\sigma^2 = \bar{x}^2$ (with $\bar{x} > 0$).

$\bar{x} = \dfrac{53 + x}{4}$ and $\sum x_i^2 = 169 + 361 + 441 + x^2 = 971 + x^2$, so $\sigma^2 = \dfrac{971 + x^2}{4} - \bar{x}^2$.

$16\sigma^2 = \bar{x}^2 \Rightarrow 4(971 + x^2) = 17\bar{x}^2 = \dfrac{17(53 + x)^2}{16} \Rightarrow 64(971 + x^2) = 17(53 + x)^2$.

$62144 + 64x^2 = 47753 + 1802x + 17x^2 \Rightarrow 47x^2 - 1802x + 14391 = 0$.

Discriminant $= 1802^2 - 4 \cdot 47 \cdot 14391 = 541696 = 736^2$, so $x = \dfrac{1802 \pm 736}{94}$, i.e. $x = 27$ or $x = \dfrac{533}{47}$.

As $x$ is an integer, $x = 27$.

Check: $13, 19, 21, 27$ have mean 20, deviations $-7, -1, 1, 7$, variance $\dfrac{100}{4} = 25$, $\sigma = 5$, CV = 25%.

Adding $c$ to every observation leaves $\sigma = 5$ and makes the mean $20 + c$. CV $= 40\%$: $\dfrac{5}{20+c} = \dfrac{2}{5} \Rightarrow 20 + c = 12.5 \Rightarrow c = -7.5$.

**Final Answer: -7.5**

## Question 27

**English**

The four observations $a,\ a+d,\ a+2d,\ a+3d$, where $d > 0$, have arithmetic mean 11. In a frequency distribution, the values $a,\ a+d,\ a+2d,\ a+3d$ occur with frequencies $1, 3, 9, 27$, respectively, and the arithmetic mean of this frequency distribution is 15.2. Then the variance of this frequency distribution is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

**हिन्दी**

चार प्रेक्षणों $a,\ a+d,\ a+2d,\ a+3d$, जहाँ $d > 0$ है, का समान्तर माध्य 11 है। एक बारंबारता बंटन में मान $a,\ a+d,\ a+2d,\ a+3d$ क्रमशः बारंबारताओं $1, 3, 9, 27$ के साथ आते हैं, तथा इस बारंबारता बंटन का समान्तर माध्य 15.2 है। तब इस बारंबारता बंटन का प्रसरण _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

**Solution**

The mean of the four AP terms is the mean of the first and last: $\dfrac{a + (a+3d)}{2} = a + \dfrac{3}{2}d = 11$.

In the frequency distribution, value $a + kd$ has frequency $3^k$ $(k = 0, 1, 2, 3)$, $N = 1 + 3 + 9 + 27 = 40$.

Mean $= a + d \cdot \dfrac{\sum k3^k}{40}$. The arithmetico-geometric sum $S = \sum_{k=0}^{3} k3^k$: $S - 3S = 3 + 9 + 27 - 3 \cdot 81 = -204$, so $S = 102$.

Mean $= a + \dfrac{102}{40}d = a + 2.55d = 15.2$.

Subtracting $a + 1.5d = 11$: $1.05d = 4.2$, so $d = 4$ and $a = 11 - 6 = 5$. Values $5, 9, 13, 17$.

Variance by coding $u = k = \dfrac{x-5}{4}$: $\sum fu = 102$, $\sum fu^2 = 0 + 3 + 36 + 243 = 282$.

$\sigma_u^2 = \dfrac{282}{40} - \left(\dfrac{102}{40}\right)^2 = 7.05 - 6.5025 = 0.5475$.

Variance of $x$: $4^2 \times 0.5475 = 8.76$. Check directly: $\sum fx^2 = 25 + 243 + 1521 + 7803 = 9592$, $\dfrac{9592}{40} - 15.2^2 = 239.8 - 231.04 = 8.76$.

**Final Answer: 8.76**

## Question 28

**English**

Two frequency distributions $A$ and $B$ take the values $10, 20, 30, 40, 50$ with the following frequencies, where $k$ is a positive integer.

| Value | 10 | 20 | 30 | 40 | 50 |
|---|---|---|---|---|---|
| Frequency in $A$ | 1 | 3 | 3 | 1 | 2 |
| Frequency in $B$ | 2 | $k$ | 1 | 3 | 5 |

The distributions $A$ and $B$ have the same arithmetic mean. If $\sigma_A^2$ and $\sigma_B^2$ denote their variances,

then the value of $\frac{\sigma_A^2}{\sigma_B^2}$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

**हिन्दी**

दो बारंबारता बंटन $A$ तथा $B$ मान $10, 20, 30, 40, 50$ निम्न बारंबारताओं के साथ लेते हैं, जहाँ $k$ एक धनात्मक पूर्णांक है।

| Value | 10 | 20 | 30 | 40 | 50 |
|---|---|---|---|---|---|
| Frequency in $A$ | 1 | 3 | 3 | 1 | 2 |
| Frequency in $B$ | 2 | $k$ | 1 | 3 | 5 |

बंटनों $A$ तथा $B$ का समान्तर माध्य समान है। यदि $\sigma_A^2$ तथा $\sigma_B^2$ इनके प्रसरण हैं, तो $\frac{\sigma_A^2}{\sigma_B^2}$ का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

**Solution**

Code $u = \dfrac{x-30}{10}$, so $u = -2, -1, 0, 1, 2$.

For $A$ ($N = 10$): $\sum fu = -2 - 3 + 0 + 1 + 4 = 0$, so the mean of $A$ is 30.

For $B$ ($N = 11 + k$): $\sum fu = -4 - k + 0 + 3 + 10 = 9 - k$. Equal means require $\sum fu = 0$, so $k = 9$ and $N = 20$.

$A$: $\sum fu^2 = 4 + 3 + 0 + 1 + 8 = 16$, $\sigma_u^2 = 1.6$, $\sigma_A^2 = 160$.

$B$: $\sum fu^2 = 8 + 9 + 0 + 3 + 20 = 40$, $\sigma_u^2 = 2$, $\sigma_B^2 = 200$.

$\dfrac{\sigma_A^2}{\sigma_B^2} = \dfrac{160}{200} = 0.8$. With equal means, $A$ is the more consistent distribution.

**Final Answer: 0.8**

## Question 29

**English**

Consider the five observations $1, 2, 8, 10, 14$. Two of these observations are replaced by two integers $x$ and $y$ with $x < y$, in such a way that the mean of the five observations is unchanged and their standard deviation becomes twice the original standard deviation. Then the value of $x$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

**हिन्दी**

पाँच प्रेक्षणों $1, 2, 8, 10, 14$ पर विचार कीजिए। इनमें से दो प्रेक्षणों को दो पूर्णांकों $x$ तथा $y$, जहाँ $x < y$ है, से इस प्रकार प्रतिस्थापित किया जाता है कि पाँचों प्रेक्षणों का माध्य अपरिवर्तित रहता है तथा उनका मानक विचलन मूल मानक विचलन का दुगुना हो जाता है। तब $x$ का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

**Solution**

Original mean $= \dfrac{35}{5} = 7$; deviations $-6, -5, 1, 3, 7$; $\sum(x_i - 7)^2 = 120$, so $\sigma^2 = 24$.

Let $p, q$ be the replaced observations. The mean is unchanged, so $x + y = p + q$.

With the same mean, $5\sigma^2 = \sum x_i^2 - 5 \cdot 49$. The new variance is $4 \cdot 24 = 96$, so the sum of squares increases by $5(96 - 24) = 360$: $x^2 + y^2 = p^2 + q^2 + 360$.

Then $(y - x)^2 = 2(x^2 + y^2) - (x + y)^2 = 2(p^2 + q^2) + 720 - (p + q)^2 = (p - q)^2 + 720$.

For integers $x, y$, $(p-q)^2+720$ must be a perfect square. The possible $|p-q|$ are $1, 7, 9, 13, 6, 8, 12, 2, 6, 4$;

the values $(p-q)^2+720$ are $721, 769, 801, 889, 756, 784, 864, 724, 756, 736$, and only $784 = 28^2$ is a square, for $\{p, q\} = \{2, 10\}$.

Then $x + y = 12$ and $y - x = 28$, so $x = -8$ and $y = 20$ (both integers).

Check: $1, 8, 14, -8, 20$ have mean 7 and $\sum(x_i - 7)^2 = 36 + 1 + 49 + 225 + 169 = 480$, variance 96, standard deviation $4\sqrt{6} = 2 \cdot 2\sqrt{6}$.

**Final Answer: -8**